

① How would the stopping potential for a given photosensitive surface change if (i) the frequency of the incident radiation were increased? and (ii) the intensity of incident radiation were decreased? Justify your answer.

② Identify the electromagnetic wave whose wavelength range is from a) 10^{-12}m to 10^{-8}m
b) 10^{-3}m to 10^1m .

Write one use of each.

③ An electric dipole of length 2cm is placed at an angle of 30° with an electric field $2 \times 10^5 \text{ N/C}$ if the dipole experiences a Torque of $8 \times 10^{-3} \text{ Nm}$ then the magnitude of either charge of the dipole is _____?

④ A square shaped coil of side 10cm , having 100 turns is placed perpendicular to a magnetic field which is increasing at the rate 1 T/s . The induced emf in the coil is _____.

⑤ In YDSE, the screen is moved away from the plane of the slits. What will be its effect on the following?

- (i) Angular separation of the fringes
- (ii) Fringe width.

⑥ The magnifying power of an astronomical telescope in normal adjustment is 2.9 and the objective and the eyepiece are separated by a distance of 150cm . Find the focal length of the two lenses.